Shell Scripting

Shell scripting in Linux is a way to automate a sequence of commands into a single executable file. It acts as an intermediary between the user and the Linux Kernel, enabling developers and system administrators to streamline repetitive tasks like backups, system monitoring, and software deployments.

- Interprets and validates user commands
- Converts commands into system calls
- Executes built-in and external programs
- Displays output or error messages
- Provides scripting capabilities

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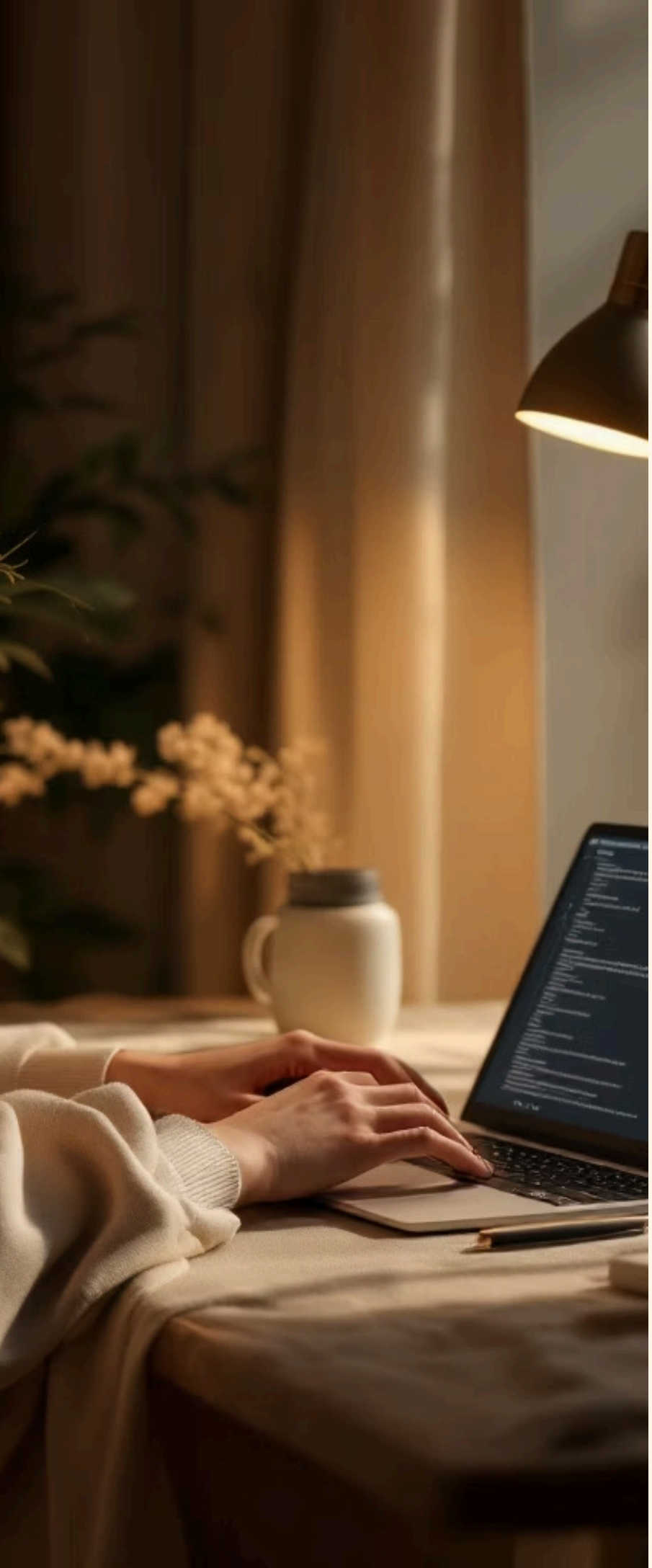

Writing Your First Bash Script

1

Create the File

Use any text editor to create a file with a `.sh` extension, e.g., `myscript.sh`.

2

Add the Shebang

Start every script with `#!/bin/bash` to tell the system which interpreter to use.

3

Write Your Commands

Add standard shell commands, variables, conditionals, or loops as needed.#!/bin/bash

echo "Hello, World!"

4

Make It Executable

Run `chmod +x myscript.sh` to grant execute permissions before running.

5

Run the Script: Execute it by typing: `./myscript.sh`

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Executing Bash Scripts

Three Ways to Run a Script

→ Direct Execution

`./myscript.sh` — runs the script in the current directory (requires `+x` permission).

→ Via Bash Interpreter

`bash myscript.sh` — no execute permission needed; bash handles it directly.

→ With Full Path

`/home/user/scripts/myscript.sh` — useful when the script lives outside your current directory.

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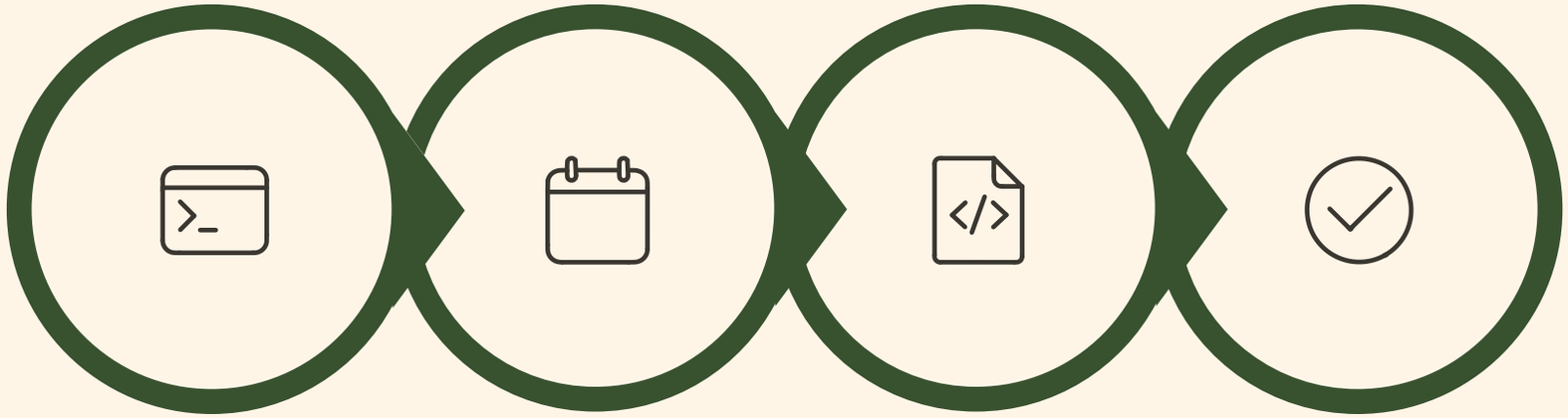
Scheduling Scripts with cron



Edit crontab

Define schedule

Specify command

Save & run

Use `crontab -e` to open your user's cron table. Each line follows the format: `* * * * * /path/to/script.sh` — representing minute, hour, day of month, month, and day of week.

Pro tip: Always use absolute paths in cron jobs. Cron runs with a minimal environment, so `/usr/bin/bash` and full script paths prevent "command not found" errors.

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Searching Files with find & locate

find — Real-Time Search

Searches the filesystem on demand with powerful filtering.

- `find /var/log -name "*.log"` — find by name
- `find . -type f -size +100M` — find large files
- `find /home -mtime -7` — modified in last 7 days
- `find . -exec chmod 644 {} \;` — execute on results

locate — Fast Database Search

Queries a pre-built database — much faster but may be slightly stale.

- `locate nginx.conf` — quick file lookup
- `locate -i "*.bak"` — case-insensitive search
- `sudo updatedb` — refresh the database
- `locate --count pattern` — count matches only

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Archiving & Compressing with tar & gzip

Create Archive

tar -cvf backup.tar /data

-c create, -v verbose, -f filename

Compress with gzip

gzip backup.tar

Or combine: tar -czvf backup.tar.gz /data

Extract Archive

tar -xzvf backup.tar.gz

-x extract, -z gzip, -v verbose

List Contents

tar -tzf backup.tar.gz

Preview files inside without extracting

Quick reference: Use .tar.gz for compressed archives (saves space) and .tar for uncompressed (faster to access). Always test extraction with tar -tzf before deploying.

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